

RECTANGULAR-CORNER STRIPPING OF INTEGER PARTITIONS: CLOCKS, FIBRES, AND PARAMETER RECOVERY

ANONYMOUS

ABSTRACT. Fix positive integers a, b . From the Ferrers diagram of an integer partition, delete the first a rows and first b columns and iterate on the translated southeast remainder. We determine this rank-changing finite dynamics on all partitions of weight at most N . A cell-coordinate formula gives every iterate, a pointwise rectangle-survival clock, and the sharp height

$$\min\{t \geq 0 : (at + 1)(bt + 1) > N\}.$$

Independently, we enumerate every rank- t fibre. For a nonempty target μ , the weight series is a forced monomial times $1/((q; q)_{at}(q; q)_{bt})$; the empty target has a separate finite hook series. This yields an exact image threshold under every weight cap. The thresholds of the one-cell, two-cell row, and two-cell column targets recover the ordered pair (a, b) . First-row/first-column deletion, its Durfee decrement, and classical partition products are explicit zero-credit inputs. A deterministic audit checks 3,462,895 exact consequences.

1. SYSTEM AND SOURCE BOUNDARY

Let $\mathcal{P}$ be the set of integer partitions, including $\emptyset$. We use English Ferrers diagrams: (i, j) is a cell of $\lambda = (\lambda_1, \lambda_2, \dots)$ when $1 \leq j \leq \lambda_i$. For fixed $a, b \geq 1$, define

$$(1) \quad T_{a,b}(\lambda) = (\lambda_{a+1} - b, \lambda_{a+2} - b, \dots)_+,$$

where nonpositive parts are omitted. Geometrically, we delete the first a rows and first b columns, then translate the southeast remainder. Since weight decreases on every nonempty state, (1) restricts to the finite carrier $\mathcal{P}_{\leq N} = \{\lambda : |\lambda| \leq N\}$.

The operation has classical inputs. Barnes and Savage delete the first row and column in a recurrence for graphical partitions and explicitly note that the Durfee size decreases by one [2]. Andrews and Eriksson give the standard Ferrers, Gaussian-polynomial, bounded-partition, and Durfee-square background [1]. We assign zero credit to those facts, to Ferrers conjugation, and to the product

$$(2) \quad \frac{1}{(q; q)_r} = \prod_{j=1}^r \frac{1}{1 - q^j}, \quad (q; q)_0 = 1,$$

which enumerates partitions with at most r parts, equivalently those whose largest part is at most r . Our scoped object is the joint two-parameter temporal and target-resolved atlas below. The source search is bounded; no novelty, priority, or owner-absence conclusion is drawn.

TABLE 1. Source subtraction and retained scope.

Record	Zero-credit input	Scoped residual
Barnes–Savage [2]	delete one row and column; Durfee decrement	all a, b, t , sharp capped height, target atlas
Andrews–Eriksson [1]	Ferrers diagrams, Durfee squares, bounded-partition products	forced-target/two-boundary inverse and exact thresholds
Generic finite maps	orbit, image, fibre terminology	formulas tied to the literal coordinate window (1)

For a nonempty μ , write $\ell(\mu)$ for its length and set, for $h, w \geq 0$,

$$(3) \quad M_{h,w}(\mu) = |\mu| + h(\mu_1 + w) + w\ell(\mu),$$

$$(4) \quad E_{h,w}(q) = \frac{1}{(q; q)_h} \sum_{k=0}^w \frac{q^{k(h+1)}}{(q; q)_k}.$$

Let $\tau_{a,b}(\lambda)$ be the least t for which $T_{a,b}^t(\lambda) = \emptyset$.

2. THE RECTANGLE CLOCK

The temporal statement is pointwise before it is enumerative.

Theorem 1 (iterates, clock, and sharp height). *For $a, b \geq 1$ and $t \geq 0$, put $h = at, w = bt$. Then*

$$(5) \quad T_{a,b}^t(\lambda) = (\lambda_{h+1} - w, \lambda_{h+2} - w, \dots)_+.$$

Consequently

$$(6) \quad \tau_{a,b}(\lambda) = \min\{t \geq 0 : \lambda_{at+1} \leq bt\}, \quad \tau_{a,b}(\lambda) > t \iff (at+1, bt+1) \in \lambda,$$

where missing parts are zero. The empty partition is the unique recurrent state, and the maximum entry time on $\mathcal{P}_{\leq N}$ is

$$(7) \quad H_{a,b}(N) = \min\{t \geq 0 : (at+1)(bt+1) > N\}.$$

Moreover, the exact weight-refined absorbed census is

$$(8) \quad \sum_{\lambda: T_{a,b}^t(\lambda) = \emptyset} q^{|\lambda|} = E_{at,bt}(q).$$

Thus $[q^n]E_{at,bt}$ counts weight- n states absorbed by rank t ; consecutive differences give every exact clock shell.

Proof. One update retains exactly the cells (i, j) satisfying $i > a, j > b$ and sends them to $(i-a, j-b)$. Induction adds these offsets and proves (5). The retained diagram is nonempty exactly when its northwest cell existed before translation, which is (6). Every nonempty state loses $(1, 1)$, so only $\emptyset$ is recurrent.

Survival through rank t forces the rectangle $[1, at+1] \times [1, bt+1]$ and therefore at least $(at+1)(bt+1)$ cells. Conversely, that rectangle itself survives rank t . This proves (7), including $H_{a,b}(0) = 0$.

It remains to count $T^t(\lambda) = \emptyset$, equivalently $\lambda_{h+1} \leq w$. Slice by $k = \lambda_{h+1}$. For $k = 0$, the diagram has at most h rows and contributes $1/(q; q)_h$. For $1 \leq k \leq w$, remove a k -by- $(h+1)$ rectangle. The excess in the first h rows is an arbitrary partition with at most h parts, while the remaining lower rows form an arbitrary partition with largest part at most k . This reversible decomposition contributes $q^{k(h+1)}/((q; q)_h(q; q)_k)$. Summing the disjoint slices proves (8). $\square$

For $a = b = 1$, (6) is the classical Durfee-size statement; that specialization is contextual rather than part of the retained claim. For unequal a, b , the clock tests a moving rectangular corner and the capped height is the first failure of a quadratic area inequality.

3. EVERY TARGET AT EVERY TIME

The clock sees one corner cell. The inverse problem instead records two entire boundaries around a prescribed southeast remainder.

Theorem 2 (complete fibre atlas). *Let $t \geq 0$, $h = at$, and $w = bt$. For every target $\mu \in \mathcal{P}$,*

$$(9) \quad \sum_{\lambda: T_{a,b}^t(\lambda) = \mu} q^{|\lambda|} = \begin{cases} \frac{q^{M_{h,w}(\mu)}}{(q; q)_h(q; q)_w}, & \mu \neq \emptyset, \\ E_{h,w}(q), & \mu = \emptyset. \end{cases}$$

For $t \geq 1$, a nonempty μ lies in the rank- t image of $\mathcal{P}_{\leq N}$ exactly when

$$(10) \quad M_{at,bt}(\mu) \leq N.$$

When this holds, μ has a source of every weight from its threshold through N ; the corresponding coefficient in (9) is the exact size-refined fibre.

Proof. The empty line is (8). Fix nonempty $\mu = (\mu_1, \dots, \mu_r)$. Formula (5) says that a source has the forced middle rows

$$(11) \quad \lambda_{h+j} = \mu_j + w \quad (1 \leq j \leq r), \quad \lambda_{h+r+1} \leq w.$$

The first h rows have baseline $\mu_1 + w$. Their excesses form an arbitrary partition γ with at most h parts. The rows below the block in (11) form an arbitrary partition β with largest part at most w .

Conversely, after padding γ with zeros, any such pair (γ, β) reconstructs a unique weakly decreasing source. The forced weight is

$$h(\mu_1 + w) + \sum_{j=1}^r (\mu_j + w) = M_{h,w}(\mu),$$

and the two independent free pieces contribute $1/(q; q)_h$ and $1/(q; q)_w$. This proves the nonempty line of (9). At $t = 0$, both orders are zero and the sole source is μ , as required.

For $t \geq 1$, both h and w are positive. The product $1/((q; q)_h(q; q)_w)$ has a positive coefficient in every nonnegative degree: one may add any number of parts of size one to γ . Hence no source exists below the forced weight and a source exists at every weight at or above it. Intersecting with $\mathcal{P}_{\leq N}$ proves (10). $\square$

The separate empty line is essential. Substituting an undefined μ_1 or $\ell(\mu)$ into the nonempty formula would miss all hook-shaped sources. Likewise, the threshold claim is restricted to $t \geq 1$; at $t = 0$, the fibre consists only of the target at its exact weight.

4. COEFFICIENT CONSEQUENCES AND ONE WORKED FIBRE

Let

$$p_r(d) = [q^d] \frac{1}{(q; q)_r},$$

with $p_r(d) = 0$ for $d < 0$. Thus $p_r(d)$ counts partitions of d with at most r parts. Reading coefficients in Theorem 2 gives the fully numerical form

$$(12) \quad \#\{\lambda \vdash n : T_{a,b}^t(\lambda) = \mu\} = \sum_{j=0}^{n-M_{h,w}(\mu)} p_h(j) p_w(n - M_{h,w}(\mu) - j)$$

for nonempty μ , where the sum is zero when $n < M_{h,w}(\mu)$. This formula separates the top and bottom freedom even after the total source weight is fixed.

For example, take $(a, b) = (2, 1)$, $t = 2$, and $\mu = (3, 1)$. Then $h = 4$, $w = 2$ and

$$M_{4,2}((3, 1)) = 4 + 4(3 + 2) + 2 \cdot 2 = 28.$$

The target therefore first appears at source weight 28, and its exact fibre series is

$$(13) \quad \frac{q^{28}}{(q; q)_4(q; q)_2} = q^{28}(1 + 2q + 5q^2 + 9q^3 + 17q^4 + \dots).$$

Hence its fibres at source weights 28, 29, 30, 31, 32 have sizes 1, 2, 5, 9, 17. The corresponding empty fibre is not obtained by setting $\mu = \emptyset$ here; by (4) it is

$$E_{4,2}(q) = \frac{1}{(q; q)_4} \left(1 + \frac{q^5}{(q; q)_1} + \frac{q^{10}}{(q; q)_2} \right).$$

Two aggregate consequences are useful checks on the atlas. First, the exact rank- t image size on the capped carrier is

$$(14) \quad 1 + \#\{\mu \neq \emptyset : M_{at,bt}(\mu) \leq N\};$$

the initial one counts the empty target. Second, because the fibres partition all sources, summing (9) over targets gives the formal mass identity

$$(15) \quad \frac{1}{(q; q)_\infty} = E_{h,w}(q) + \frac{1}{(q; q)_h(q; q)_w} \sum_{\mu \neq \emptyset} q^{M_{h,w}(\mu)}.$$

This is not an additional ownership claim; it is the global consistency equation for the target-resolved bijection.

5. DUALITY AND PARAMETER RECOVERY

Let λ' be the conjugate partition. Transposing cell coordinates in the definition immediately gives

$$(16) \quad T_{a,b}(\lambda)' = T_{b,a}(\lambda').$$

This symmetry explains why a square clock alone cannot orient the two parameters. The target thresholds do.

Corollary 3 (three-probe identifiability). *Let $m(\mu)$ denote the minimum source weight of the one-step nonempty target μ . Then*

$$(17) \quad m((1)) = (a+1)(b+1),$$

$$(18) \quad m((2)) = m((1)) + a + 1,$$

$$(19) \quad m((1,1)) = m((1)) + b + 1.$$

Therefore

$$(20) \quad a = m((2)) - m((1)) - 1, \quad b = m((1,1)) - m((1)) - 1.$$

The ordered update parameters are determined by the one-step target-support profile.

Proof. Theorem 2 identifies $m(\mu)$ with $M_{a,b}(\mu)$. Substitution of the three displayed shapes gives

$$M_{a,b}((1)) = 1 + a(1+b) + b,$$

$$M_{a,b}((2)) = 2 + a(2+b) + b,$$

$$M_{a,b}((1,1)) = 2 + a(1+b) + 2b.$$

Taking the two differences proves (17)–(20). $\square$

This interface is stronger than recovering only the symmetric product $(a+1)(b+1)$. Row and column probes retain the orientation exchanged by (16).

6. EXACT CONTROL AND LIMITATIONS

The accompanying standard-library verifier enumerates every partition through weight 32. For $(a,b) = (1,1), (2,1), (1,3), (2,2), (3,2)$ and $0 \leq t \leq 5$, it compares literal repeated updates with (5); checks every clock and capped height; checks conjugation duality; and compares every fibre coefficient for targets through weight nine with (9). It also checks fibre-mass conservation, the zero/nonzero image threshold, and all three recovery probes. The frozen run contains 3,462,895 exact assertions.

Enumeration is falsification pressure, not proof. The theorem is formal and enumerative: it makes no asymptotic statement, no mixing claim, and no claim about a probability law unless a distribution on partitions is separately chosen. The bounded owner search does not certify absence, and all external release, posting, authorship, and submission decisions remain outside this anonymous internal note.

REFERENCES

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